



E-Commerce Sales Data Analysis

End-to-end analysis using **Excel**, **MySQL**, and **Power BI** — turning raw data into business insights.

Project Overview



The Goal

Convert raw e-commerce sales data into **meaningful business insights** through:

- Structured data cleaning
- SQL-based analysis
- Interactive dashboard reporting

Data Cleaning & Preparation

EXCEL

Deduplication

Removed duplicates & null values

Standardization

TRIM, UPPER, LOWER, PROPER functions

Data Types

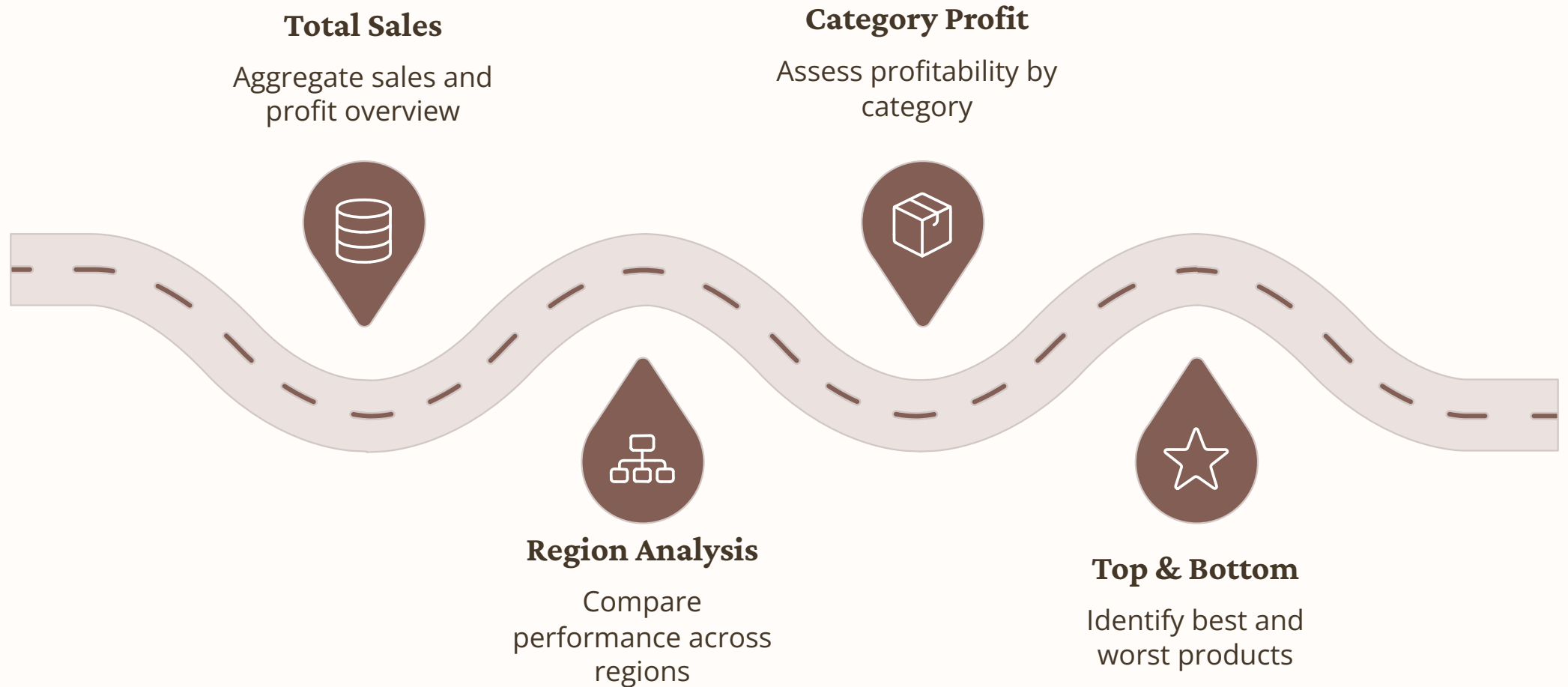
Corrected Date, Numeric formats

Pivot Analysis

Region, Category, Product breakdowns



SQL Business Analysis



Structured queries uncovered performance patterns across regions, categories, and products.

Product Performance Queries

Top Performers

- Top 5 products by sales
- Top 5 brands by revenue
- Top 3 cities by sales
- Products above average sales

Loss Makers

- Bottom 5 loss-making products
- Products below average profit
- Discount impact on margins
- Average sales per product



Power BI Dashboard

POWER BI

Cleaned MySQL data imported into Power BI. DAX measures built for **Total Sales**, **Total Profit**, **Avg. Delivery Days**, and **Order Count**.

Dashboard Features



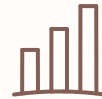
KPI Cards

At-a-glance metrics



Regional Comparison

Side-by-side region visuals



Bar Charts

Product & regional performance



Dynamic Slicers

Interactive filtering

Key Business Insights

Revenue Concentration

A **few products** generate the majority of total revenue

Discount Danger

High discounts significantly reduce profit margins

Regional Gaps

Certain regions **significantly outperform** others

Loss Sub-categories

Some sub-categories **consistently generate losses**



Recommendations

01

Optimize Discounts

Revise discount strategies to protect margins

02

Focus on High-Margin Products

Prioritize top-performing SKUs and brands

03

Review Underperformers

Audit loss-making sub-categories

04

Invest in Top Regions

Double down on high-performing markets

Future Enhancements



Next Steps

- Predictive analytics integration
- Customer segmentation modeling

□ This project demonstrates a complete **Excel → MySQL → Power BI** analytics pipeline ready for real-world business decisions.